

Testing hypothesis testing



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Abstract

This report investigates the p-values that are generated when we perform hypothesis tests by treating them as if they are random variables. We show that you can compute histograms and summary statistics by performing hypothesis tests multiple times. We then show how performing these analysis allows us to better understand the reasons we might observe false positive/negative results when we perform statistical tests.

1 Introduction

The p-value that is the final result of a hypothesis test can be thought of as a random variable just like the Bernoulli, binomial and normal random variables that we have studied in other exercises. In this report, we will use the methods of sampling that have been covered elsewhere in the module material to study the distribution of values that are obtained when we perform hypothesis tests to test whether the expectation of a sampled normal distribution has a particular value.

We begin in section 3 by calculating the test statistic by sampling from the distribution that is assumed under the null hypothesis. In section 4 we compute the test statistic by sampling from a distribution that is different from the one assumed under the null hypothesis. We show that the false negative rate is strongly affected by the effect size but also that, if sufficient sampling is performed, a statistically significant effect can be observed even if the effect size is relatively modest.

2 Methods

Throughout this report we will draw n samples, X_i , from a normal random variable with unit variance. The null hypothesis for all our tests will be that the expectation of the sampled variable is equal to zero. We will then perform one and two-sided tests to determine if the expectation for the sampled population is greater than zero or not equal to zero respectively. Our test statistic is thus computed using:

$$T = \frac{\bar{X}}{1/\sqrt{n}} \quad \text{where} \quad \bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

The p-value for the one and two sided hypothesis tests are then:

$$\text{one-sided: } p = 1 - \Phi(T) \quad \text{two-sided: } p = 2[1 - \Phi(|T|)]$$

where Φ is the cumulative distribution for the normal random variable.

3 Sampling p-values

We can sample the p-value obtained from a hypothesis test in the same way that we can sample any other random variable. For example, to construct figure 1 we performed one hundred hypothesis tests of the null hypothesis that $\mu = 0$ against an alternative hypothesis that $\mu > 0$. To sample a distribution of p-values that one would obtain from this test we generated 100 sample means each of which was computed by taking the average of 100 standard normal random variables. Figure 1 shows that the p-values from this experiment take values between 0 and 1. Furthermore, and most importantly, four of the p-values we obtain are less than 5%, which we would typically take as evidence for rejecting the null hypothesis in favor of the alternative. However, because we know what distribution was sampled here we know that these four results are false positives.

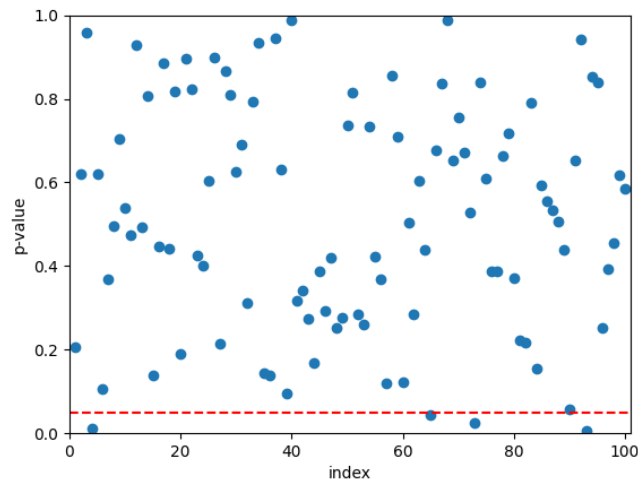


Fig. 1. A sample of p-values that were obtained from 100, one-sided hypothesis tests with the null hypothesis $\mu = 0$ and the alternative hypothesis $\mu > 0$. Each of the sample means for the 100 tests was computed from 100 normal random variables with $\mu = 0$ and $\sigma^2 = 1$. The red dashed line indicates the 5% confidence limit on the test. The four sampled p-values that fall below this line are false positives.

As you have seen in the exercises on random variables, whenever we have a sample of random variables, we can construct a histogram as well as summary statistics. To construct figure 2 we thus performed 10000 hypothesis tests of the null hypothesis that $\mu = 0$ against an alternative hypothesis that $\mu \neq 0$. To sample a distribution of p-values that one would

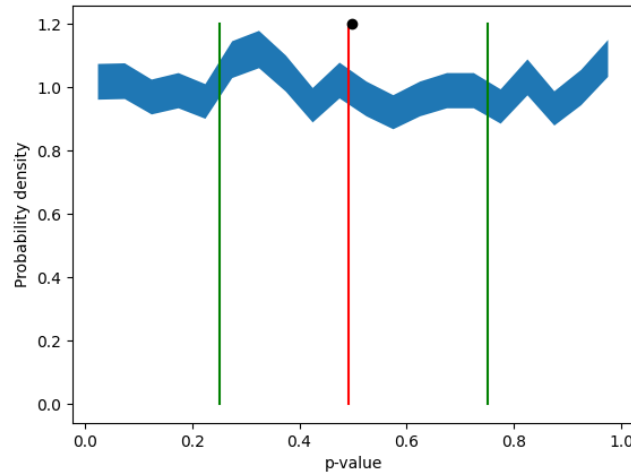


Fig. 2. An estimate for the probability density for the p-value that you would obtain from performing a hypothesis test with the null hypothesis $\mu = 0$ and the alternative hypothesis $\mu \neq 0$. Each of the sample means for the 10000 tests was computed from 100 normal random variables with $\mu = 0$ and $\sigma^2 = 1$. The blue shaded area indicates the 90% confidence limit on the estimate of the probability density function that we obtained by constructing a histogram with 20 bins. The green lines indicate the lower and upper quartile for the sample that was used to construct the histogram, while the red line is the median. The black dot is the mean. Error bars for a 90% confidence limit on this estimate for the expectation are displayed but they are smaller than the symbol used to display the mean.

obtain from this test we generated 100 sample means each of which was computed by taking the average of 100 standard normal random variables.

Figure 2 shows that the probability density function for the p-value for the two-sided hypothesis test resembles that of a continuous uniform distribution with $a = 0$ and $b = 1$. This result makes sense as the variable sampled here is given by:

$$\bar{X} = \frac{1}{100} \sum_{i=1}^{100} X_i$$

where X_i is a standard normal random variable. This random variable should be a sample from a normal distribution with $\mu = 0$ and $\sigma = 1/\sqrt{100} = 1/10$, which is precisely the distribution that our null hypothesis is testing for. Consequently, even though the test statistics that we obtain by sampling are normally distributed and not uniformly distributed, the p-values that are computed by inserting the values of the test statistic that we obtain from sampling into the cumulative distribution for the normal random variable they are sampled from are uniformly distributed.

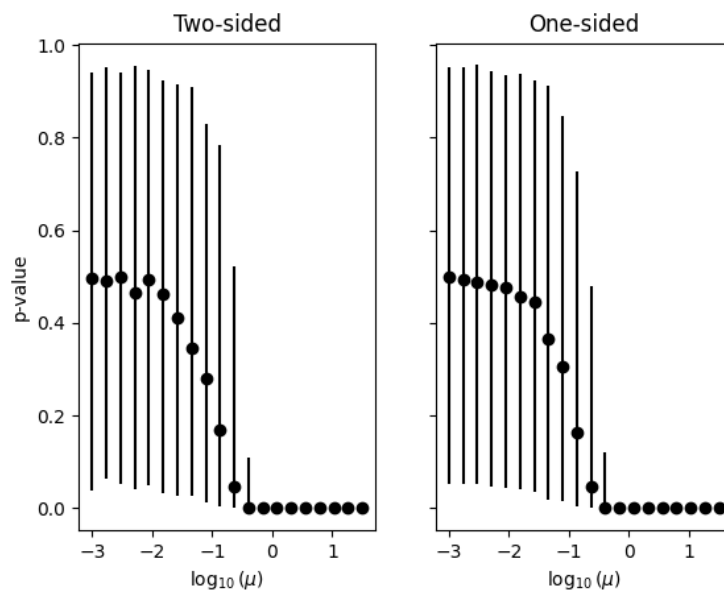


Fig. 3. Representations of the distribution p-values that were obtained from 1000, two-sided (left panel) and one-sided (right panel) hypothesis tests at each effect size. In all these tests the null hypothesis is $\mu = 0$, while the alternative hypothesis for the two-sided and one-sided tests are $\mu \neq 0$ and $\mu > 0$ respectively. Each of the sample means for the 1000 tests were computed from 50 normal random variables with the μ values indicated on the x-axis and with $\sigma^2 = 1$. Black dots indicate the median from the distribution of p-values, while the error bar extends from the 5th percentile of the distribution to the 95 percentile.

4 Exploring false negatives

In the previous section we explored the distribution of p-values that is obtained when samples are taken from the distribution that is assumed under the null-hypothesis. In this section we will explore the distribution of p-values that is obtained when we sample from a distribution that is different from the one that is assumed under the null hypothesis. In other words, we will study how having a non-zero effect size affects the p-values we sample.

Figure 3 shows how the sampled p-values obtained by doing repeated hypothesis test depend on the effect size. For all these tests the null hypothesis was that samples were generated from a distribution with $\mu = 0$. The alternative hypothesis for the one and two sided tests were then $\mu > 0$ and $\mu \neq 0$ respectively. For each value of μ on the x-axis we generated 1000 sample means each of which was computed from 50 normal random variables drawn from $N(\mu, 1)$. You can see that when μ is large, and the effect is large, the p-values are small and that we thus have sufficient evidence to reject the null hypothesis in favor of the alternative. However, when μ is small, and the effect is small, there is often insufficient evidence to reject the null hypothesis as the p-values are large. For these small effect sizes we thus frequently obtain a false negative result from our hypothesis test.

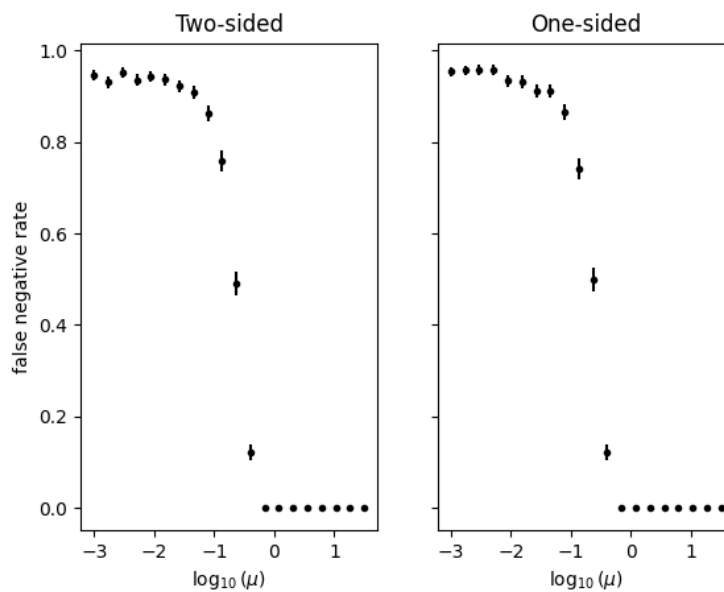


Fig. 4. False negative rates for the hypothesis tests that were performed to construct figure 3. The error bars in these figure indicate the 90% confidence limit on the estimate of the rate that is obtained by sampling.

You can more clearly see this propensity for false negative results when μ is small from figure 4, which shows the fraction of false negative results (p-values of more than 0.05) in each of our 1000 tests. Furthermore, the fact that you get a false negative result when $\mu \approx 0$ makes intuitive sense. When the difference between the mean of the sampled distribution and the mean of the distribution that is assumed under the null hypothesis is small you need a lot of data to confirm that any observed difference is the mean is not due to random fluctuations.

We can confirm that a having more data provides us with more confidence that any result that we obtain is not a false negative by constructing figures like that shown in the bottom panels of figure 5. These figure were constructed by performing 500 tests of the null hypothesis $\mu = 0$ against either $\mu \neq 0$ (two-tailed) or $\mu > 0$ (one tailed). We generated each of the sample means that were used in these tests by taking the number of samples indicated on the x -axis in the figure from a normal distribution with $\mu = 0.5$ and $\sigma = 1$. Obviously, the distribution that we are sampling from here **does not** have $\mu = 0$. However, when the sample size the means are computed from is small many of our tests give a false negative result in which the hypothesis test returns a p-value that is greater than 0.05.

For completeness figure 5 also shows the false positive rate that we obtain when the means are calculated by sampling a distribution with $\mu = 0$. You can see that the false positive rate is always close to the red dashed line that shows the expression $y = 0.05$, which is what would be expected. We know from figure 2 that the p-value we obtain when we sample from the distribution that is assumed under the null hypothesis is a uniform continuous random variable that lies between 0 and 1. We would thus expect it to take a value that is less than 0.05 about 5% of the time.

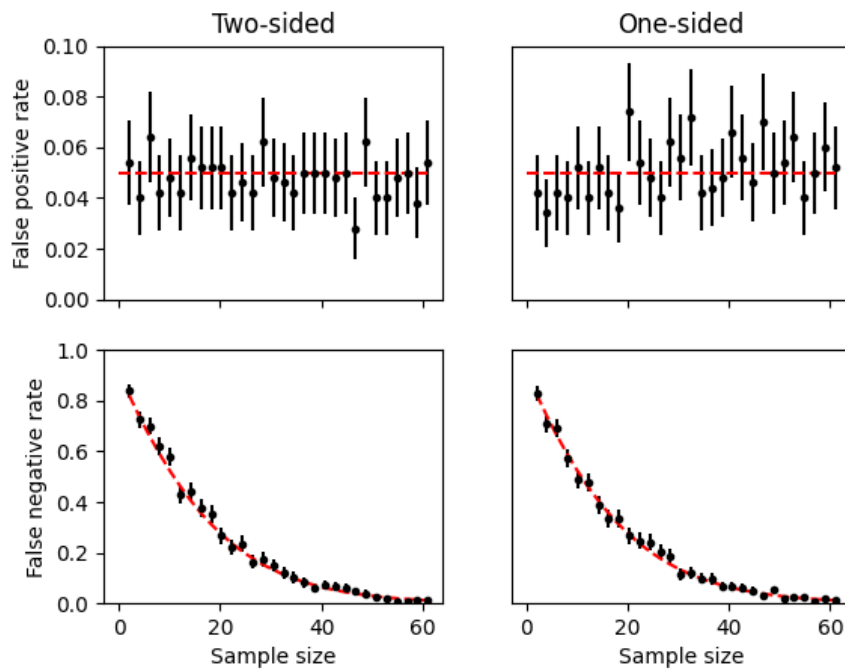


Fig. 5. False negative and false positive rates obtained when 500 two-sided and one-sided hypothesis tests were performed using different numbers of samples to calculate the mean. In all these tests a 5% significance level was used, the null hypothesis was $\mu = 0$, while the alternative hypothesis for the two-sided and one-sided tests were $\mu \neq 0$ and $\mu > 0$ respectively. For the top two panels the means were calculated by sampling a normal random variable with $\mu = 0$ so if the returned p-value is less than 0.05 the test has returned a false positive. For the bottom two panels means were calculated by sampling a normal random variable with $\mu = 0.5$ so if the returned p-value is greater than 0.05 the test has returned a false negative. The error bars indicate the 90% confidence limits on the estimates of the false positive/negative rates and red dashed lines are the expected values that these rates should take based on the theory discussed in the text.

We can derive an expression for the probability of getting a false negative result in our hypothesis test when $\mu = 0.5$. For the one-tailed test, the probability of getting a negative result can be derived as follows:

$$\begin{aligned}
 P(\text{negative hypothesis test}) &= P\left(\frac{\bar{X}}{(1/\sqrt{n})} \leq \Phi^{-1}(0.95)\right) \\
 &= P\left(\frac{\bar{X} - \mu}{(1/\sqrt{n})} \leq \Phi^{-1}(0.95) - \frac{\mu}{(1/\sqrt{n})}\right) \\
 &= \Phi\left(\Phi^{-1}(0.95) - \frac{\mu}{(1/\sqrt{n})}\right)
 \end{aligned} \tag{1}$$

The derivation above relies on the fact that the test statistic, $\frac{\bar{X}}{(1/\sqrt{n})}$, is a sample from a standard normal random variable if $\mu = 0$. We thus know we will observe a negative hypothesis test if the test statistical is less than the 95th percentile of the cumulative distribution for a standard normal random variable. To arrive at the expression in the second line we subtract $\frac{\mu}{(1/\sqrt{n})}$ from both sides of the inequality inside the probability measure. We then finish the derivation by recalling that $\frac{\bar{X} - \mu}{(1/\sqrt{n})}$ is still a standard normal random variable so we can extract the probability that we desire from the cumulative distribution for the standard normal random variable, Φ .

Using similar reasoning we can arrive at the following expression for the probability of a false negative result in the two sided test:

$$\begin{aligned}
 P(\text{negative hypothesis test}) &= P\left(\frac{\bar{X}}{(1/\sqrt{n})} \leq \Phi^{-1}(0.975)\right) - P\left(\frac{\bar{X}}{(1/\sqrt{n})} \leq \Phi^{-1}(0.025)\right) \\
 &= P\left(\frac{\bar{X} - \mu}{(1/\sqrt{n})} \leq \Phi^{-1}(0.975) - \frac{\mu}{(1/\sqrt{n})}\right) \\
 &\quad + P\left(\frac{\bar{X} - \mu}{(1/\sqrt{n})} \leq \Phi^{-1}(0.025) - \frac{\mu}{(1/\sqrt{n})}\right) \\
 &= \Phi\left(\Phi^{-1}(0.975) - \frac{\mu}{(1/\sqrt{n})}\right) - \Phi\left(\Phi^{-1}(0.025) - \frac{\mu}{(1/\sqrt{n})}\right)
 \end{aligned} \tag{2}$$

The expressions that were derived in equations 1 and 2 are shown plotted with a red dashed line in the lower panels of figure 5. You can see that there is good agreement between the results that we obtain from sampling and the predictions of these simple analytic models.